

TECH STACK DOCUMENTATION

Technology Stack & Architecture Decisions

This document outlines the core technologies used to build the AI News Explainer and the reasoning behind each choice.

Core Intelligence

Language: Python 3.10+

Why? The undisputed king of AI engineering. Massive ecosystem for LLMs (ollama), Data Science (pandas), and Media Processing (moviepy).

LLM Interaction: Ollama (Llama 3)

Why?

- * Privacy & Cost: Runs locally on the user's machine. No API fees, no data leaks.
- * Speed: Faster latency for iterative tasks like summarization compared to cloud APIs.

Architecture: Agentic Workflow (LangGraph)

Why?

- * Structured State: We use LangGraph's StateGraph to manage the flow of data between agents. This ensures type safety and predictable state transitions.
- * Modularity: Instead of one giant script, we split the brain into independent nodes (ScriptWriter, VisualPlanner, VideoComposer).
- * Resilience: If a node fails, the state graph preserves the context, allowing for potential retries.

Media Generation

Video Engine: MoviePy v2.1.1 (Beta)

Why?

- * Programmatic Control: Unlike FFmpeg (which is purely command-line), MoviePy allows pythonic manipulation of clips (composition, filters, audio).
- * v2.1.1 Choice: We chose the latest Beta despite potential instability because it offers significantly better Audio Mixing and Effect capabilities.

Image Generation: Pollinations.ai (API)

Why?

- * No Login/Key Required: zero-friction setup for the user.
- * Speed: Generates images in seconds.
- * Flexibility: Supports detailed prompts ("Unreal Engine 5 style", "No Text") which were critical for our quality upgrades.

Voice Integration: Edge-TTS

Why?

- * Neural Quality: Uses Microsoft's Azure Neural voices (e.g., en-US-EricNeural) which sound near-human, unlike the robotic standard TTS.
- * Free: Accessible without an Azure API key via the python wrapper.

Reliability & Polish

Fallback Systems

Why?

- * VisualAssetAgent: I implemented a 5-layer fallback (Real Image -> AI Gen -> AI Gen Simple -> Backup Real -> Text). This ensures the user always gets a visual response.

Strict Formatting

Why?

- * Visual Planner: Explicitly forbidding text in AI images solved the "gibberish text" hallucination problem common in image models.

Agents & Orchestration

The system uses a Linear Chain of Responsibility pattern. Each agent performs a specialized task and passes the "State" to the next.

1. NewsCollectionAgent

- * Role: The Researcher.